

Pan-Cancer Hierarchical Spike-and-Slab — Replication & Validation

Pan-Cancer Hierarchical Spike-and-Slab — Replication & Structural-Prior Validation

Pre-registered held-out validation in which structural priors derived from external biological hierarchies replace the per-cancer fitted PIPs for four covariates of the Lock-lab pan-cancer survival spike-and-slab model. Multiple papers in the corpus; each pre-registered, each with locked decision rules.

Papers in this repository

paper	hierarchy/spec	verdict
Paper 1	epithelial-class + germ-layer, 3-cov primary post-disposition (covariates 1.1, 8.2 @ epi/2a, 20.1 @ germ/3b)	MATCHES with parameter reduction — $\Delta +4.84$ nats $[-0.82, +10.05]$; 75% free-parameter reduction on the three structural-prior covariates
Paper 1 sensitivity	4-cov original pre-reg (adds cov 10.1)	nominally $\Delta +6.26$ $[+0.34, +12.29]$, but cov 10.1’s marginal contribution CI brackets zero (CHECK 1 + CHECK 2); preserved as sensitivity, not headline. See LOCK_DIAGNOSTIC_DISPOSITION.md
Paper 1 secondary	epithelial-class + germ-layer (alt hierarchy for cov 8.2)	inconclusive ($+4.33$ $[-1.86, +10.97]$)
Paper 2	operator-composition extension (18 specs)	FALSIFIES per pre-reg; L1 retro finds cancer identity = 89% of variance (specs are 18 noisy estimates of one per-cancer profile). See EXTENSION_VALIDATION_RESULTS.md
Paper 3	MOFA-FLEX niche joint refit	FALSIFIED — substrate boundary identified
Paper 4 Phase 1 (2026-05-15)	3 new external hierarchies: Thorsson immune, Malta stemness, Sanchez-Vega pathway	PAPER1_HIERARCHY_SPECIFIC — 0/3 outperform at locked CI threshold; 2/3 trend positive but statistical power insufficient at $n_test=1,362$

Paper 1 TL;DR (post-disposition headline)

The pre-registered validation was committed at 4 covariates (10.1, 1.1, 8.2 @ epi/2a, 20.1 @ germ/3b). It returned $\Delta +6.26$ nats with bootstrap 95% CI excluding zero, meeting the locked OUTPERFORMS criterion in isolation. Pre-registered post-hoc marginal-contribution diagnostics on each of the four covariates ([LOCK_DIAGNOSTIC_PRE_REGISTRATION.md](#), [LOCK_DIAGNOSTIC_DISPOSITION.md](#)) found cov 10.1's individual contribution to held-out LPPD statistically indistinguishable from zero — both CHECK 1 (per-cancer LPPD contribution: 15/14 split, 0 cancers cross $|1 \text{ nat}|$, binomial $p=1.000$) and CHECK 2 (patient-level bootstrap of cov 10.1 marginal: observed $+1.42$ nats, 95% CI $[-1.66, +4.40]$, brackets zero) return Reading A. The 3-cov primary spec dropping cov 10.1 is the headline result.

model	held-out total LPPD	Δ vs baseline	bootstrap 95% CI	verdict
baseline (Samorodnitsky/Hoadley/Lock 2022, fit on 80% training)	-1038.69	—	—	—
3-cov primary (post-disposition headline)	-1033.85	+4.84 nats	[-0.82, +10.05]	MATCHES with parameter reduction
4-cov sensitivity (original pre-reg)	-1032.43	+6.26 nats	[+0.34, +12.29]	nominally outperforms in isolation but cov 10.1 marginal CI brackets zero — see LOCK_DIAGNOSTIC_DISPOSITION.m
secondary structural-prior	-1034.36	+4.33 nats	[-1.86, +10.97]	inconclusive

Paper 4 Phase 1 TL;DR (new 2026-05-15)

Pre-registration locked at commit `677406a` before any Phase 1 Gibbs run (see [PRE_REGISTRATION_PAPER4_PHASE1.md](#)). Tests whether Paper 1's $+6.26$ nat result generalizes to three other externally-published pan-cancer hierarchies, none derived from BIDIFAC+ blocks (so no circularity flag like Hoadley supercluster).

Hierarchy	Source	Observed Δ vs baseline	95% CI	Verdict
Thorsson 2018 immune subtype (C1-C5)	Immunity 48(4):812	+4.69	[-2.37, +11.45]	INCONCLUSIVE
Malta 2018 stemness tertile (mDNasi)	Cell 173:338	+0.71	[-5.57, +7.38]	MATCHES
Sanchez-Vega 2018 dominant oncogenic pathway	Cell 173:321	+3.15	[-4.10, +9.35]	INCONCLUSIVE

Joint disposition per pre-reg: **PAPER1_HIERARCHY_SPECIFIC** (0/3 outperform, 0/3 underperform). Honest reading: two of three (Thorsson, Sanchez-Vega) directionally replicate at +47% and +50% of Paper 1's magnitude respectively, but the $n_{\text{test}}=1,362$ bootstrap CIs are $\sim \pm 7$ nats wide so neither cleanly excludes zero. Malta is genuinely near zero (+0.71). Phase 2 candidates: k-fold extension to tighten CIs, or a Bayesian probability-of-improvement decision rule. See [PHASE1_RESULTS.md](#) for full numbers + commentary.

Held-out test set: 1,362 patients across 29 cancer types, stratified random 20% per cohort, seed 42. Four chains $\times$ 100,000 Gibbs iters per model, 50k burn, 500 thinned posterior samples for LPPD evaluation, $B=1000$ patient-level bootstrap.

Decision rule (locked before any validation Gibbs run, see [PRE_REGISTRATION.md](#)): a model “outperforms” the baseline if held-out LPPD exceeds the baseline by $> +2$ nats *and* the bootstrap 95% CI on the difference excludes zero; “matches” if $|\Delta| \leq 2$ *or* the CI brackets zero. The post-disposition 3-cov primary returns $\Delta +4.84$ nats with CI bracketing zero $\rightarrow$ matches with parameter reduction. The 4-cov sensitivity meets the “outperforms” criterion in isolation but the cov 10.1 marginal-contribution diagnostic (see TL;DR above) reframes it as a sensitivity row.

The 3-cov primary structural prior replaces 87 free per-cancer beta parameters with 22 group-shared betas across the three target covariates (a 75% reduction). Held-out predictive performance is statistically indistinguishable from baseline under this constraint — consistent with productive regularization.

Provenance

The replicated method:

Samorodnitsky, S., Hoadley, K. A., & Lock, E. F. (2022). A Hierarchical Spike-and-Slab Model for Pan-Cancer Survival Using Pan-Omic Data. BMC Bioinformatics 23(1):235.
Source: https://github.com/sarahsamorodnitsky/HierarchicalSS_PanCanPanOmics

The BIDIFAC+ inputs:

Park, J., Lock, E. F., & Hoadley, K. A. (2022). Bidimensional Linked Matrix Factorization for Pan-Omics Pan-Cancer Analysis. *Annals of Applied Statistics* 16(1):484-501.

The 29 TCGA cancer types and pre-computed `XYC_V2_WithAge_StandardizedPredictors.rda` inputs come directly from the upstream repository. No re-fit of BIDIFAC+ was performed.

What's in this repo

```
.
├── README.md                ← this file
├── README.pdf              ← printable version
├── LICENSE                 ← MIT (this work) + upstream attribution
├── PRE_REGISTRATION.md     ← Paper 1 – locked decision rules
├── PRE_REGISTRATION_PAPER2.md ← Paper 2 – operator-composition extension (pre-r
├── EXTENSION_VALIDATION_RESULTS.md ← Paper 2 – 18-spec verdict + per-spec table
├── PRE_REGISTRATION_PAPER3_MOFAFLEX_NICHE.md ← Paper 3 – MOFA-FLEX niche joint-refit (pre-regi
├── MOFAFLEX_NICHE_JOINT_REFIT_RESULTS.md ← Paper 3 – paired-Δ table + verdict
├── REPLICATION_RESULTS.md  ← baseline replication numbers
├── SCREENING_SUMMARY.md    ← screening pass output
├── FALSIFICATION_REPORT.md ← 6-check adversarial audit of screening
├── VALIDATION_RESULTS.md   ← pre-registered held-out validation (Paper 1)
├── WRONGLY_GROUPED_DIAGNOSTIC.md ← within-group sign-agreement diagnostic (Paper 2)
├── COV_10_1_INVESTIGATION.md ← 3-cov sensitivity test on Paper 1 primary (Read
├── FINER_RESOLUTION_RESULTS.md ← cov 8.2 tissue-within-group ANOVA (discovery, A
├── reference/
│   └── cancer_type_hierarchies_2026-05-09.md ← 29 cancers × 5 hierarchies × granularities
├── scripts/
│   ├── replication/        ← R Gibbs runner + diagnostics
│   ├── screening/         ← Python ANOVA + falsification cascade
│   └── validation/        ← R modified-Gibbs + held-out evaluation
├── results/
│   ├── replication/       ← per-cancer × per-cov posterior table + diagnost
│   ├── screening/        ← screening + 6 falsification CSVs
│   ├── validation/       ← LPPD, bootstrap, per-cancer, convergence
│   └── mofaflex_niche_joint_refit/ ← Paper 3 paired-Δ + drift + diagnostic CSVs
└── data/
    └── README.md          ← pointers to XYC, TCGA, Zenodo (chain RDAs)
```

The chain RDA files (12 chains × ~580 MB = ~7 GB total) are **not in this repository**. They will be deposited at Zenodo with a citable DOI; the deposit URL goes here:

Zenodo DOI: TBD (will be filled in after the public release of this repo)

Without the chain RDAs, downstream analyses (convergence, held-out LPPD, bootstrap) cannot be recomputed from saved posteriors directly. The full pipeline can still be re-run from scratch using the scripts here in ~2 hours wall on a modern workstation. See `data/README.md`.

What was done, briefly

1. Replication baseline

Four-chain replication of `HierarchicalLogNormalSpikeSlab` (the upstream sampler, unmodified) on the full pre-computed XYZ inputs. R-hat max 1.0152 across 137 monitored params; per-chain σ^2 posteriors agree to 3 decimals. Cancer-specific intercepts span THCA 10.28 (≈ 80 yr at standardized predictors=0) to MESO 6.29 (≈ 1.5 yr) — clinically expected order. Pan-cancer age effect $\beta_{0.5} = -0.316$ (PIP 0.91). See `REPLICATION_RESULTS.md`.

2. Structural-hierarchy screening

ANOVA of per-cancer PIPs against five candidate hierarchies (tissue origin, epithelial classification, embryonic germ layer, sex composition, Hoadley 2018 supercluster/iCluster) at multiple granularities each. 636 tests total across 68 BIDIFAC+ covariates. 16 nominal hits at $\eta^2 > 0.30$ with $p < 0.05$. See `SCREENING_SUMMARY.md`.

3. Falsification cascade (six adversarial checks)

1. Permutation null on η^2 (B=1000): observed at chance rate
2. High-resolution perm on survivors (B=10000): 0/16 pass Bonferroni
3. BIDIFAC+ block-presence confound: Hoadley signal $\rho = +0.60$ with block presence (circular)
4. Granularity-nesting null on Spearman ρ : tissue origin marginal ($p = 0.042$), all others fail
5. Cohort-size partialing: most survivors lose $< 0.06 \eta^2$, one loses 0.15
6. Independence: 16 survivors collapse to 5 effective dimensions; 32/120 pairs Jaccard > 0.5

No single (cov, hierarchy, granularity) cell is publishable as standalone structural evidence at conventional rigor. See `FALSIFICATION_REPORT.md`.

4. Pre-registered validation

The screening result motivated a different question: do the four largest-effect-size candidates *jointly* improve held-out predictive performance when used as structural priors, even though no single candidate is individually Bonferroni-significant? Pre-registered in `PRE_REGISTRATION.md`, ran the modified Gibbs sampler at `scripts/validation/sampler_structural_prior.R`, evaluated held-out LPPD on the 20% test set with B=1000 bootstrap CI. Original 4-cov result: $\Delta = +6.26$ nats with CI excluding zero (nominal OUTPERFORMS). Post-hoc marginal-contribution diagnostics ([LOCK_DIAGNOSTIC_DISPOSITION.md](#)) found cov 10.1's individual contribution indistinguishable from zero; the 3-cov primary spec without cov 10.1 returns $\Delta = +4.84$ nats with CI bracketing zero $\rightarrow$ matches with 75% parameter reduction. The 3-cov spec is the post-disposition headline; the 4-cov spec is preserved as a sensitivity row. See `VALIDATION_RESULTS.md` and `LOCK_DIAGNOSTIC_DISPOSITION.md`.

5. Follow-up investigations (post-shipping)

Three diagnostic / sensitivity analyses, all data-only reports without disposition recommendations:

- `WRONGLY_GROUPED_DIAGNOSTIC.md` — within-group sign-agreement scan of per-cancer baseline betas across all candidate (cov × hierarchy) cells used in Paper 1 and Paper 2. Two cells flag (both cov 10.1 at epi/2a and germ/3b); cov 8.2 is clean across all four candidate hierarchies. Prerequisite (b) of the Paper 2 pre-registration.
- `COV_10_1_INVESTIGATION.md` — refit Paper 1 primary on three covariates (drop cov 10.1) to test whether cov 10.1's contribution matters. 3-cov vs 4-cov $\Delta = -1.42$ nats, bootstrap CI includes zero (Reading A holds). The 3-cov spec alone has a CI that includes zero against baseline, so cov 10.1 is the marginal covariate carrying the 4-cov result across the CI-excludes-zero threshold.
- `FINER_RESOLUTION_RESULTS.md` — ANOVA + permutation null test of whether tissue-within-group or hybrid tissue+subtype labels organize cov 8.2 residuals beyond the coarse epi/2a 3-group prior used in Paper 1. Epithelial group $n=21$, $\eta^2=0.608$ vs perm null median 0.381, $p_{\text{emp}}=0.117$. Directional consistency but not certified at this n.

How to reproduce

Two paths:

Fast path — verify reported numbers from CSV outputs

The small CSV outputs in `results/` are sufficient to regenerate every table in the four reports without rerunning any Gibbs sampler. Inspect `results/validation/loglik_summary.csv`, `bootstrap_ci_summary.csv`, `per_cancer_loglik_wide.csv`, etc. directly.

Full path — re-fit everything from raw data

~2 hours wall on a modern workstation, four parallel chains per Gibbs run, three Gibbs runs sequentially. Step-by-step instructions in `data/README.md`. Briefly:

```

# 1. Pre-computed XYC inputs (clone upstream repo)
git clone https://github.com/sarahsamorodnitsky/HierarchicalSS_PanCanPanOmics.git

# 2. Replication baseline (4 chains × 100k iters; ~36 min wall)
for c in 1 2 3 4; do
  Rscript scripts/replication/run_gibbs_chain.R $c &
done
wait
Rscript scripts/replication/diagnostics.R
Rscript scripts/replication/per_cancer_summary.R

# 3. Screening + 6 falsification checks (~30 min)
python scripts/screening/hierarchies_long.py
python scripts/screening/run_screening.py
python scripts/screening/falsify_0{1,2,3,4,5,6}_*.py
python scripts/screening/synthesize_summary.py

# 4. Pre-registered validation (3 Gibbs runs sequentially × 4 parallel chains; ~2 h)
Rscript scripts/validation/01_build_split_and_targets.R
for spec in baseline_train primary secondary; do
  for c in 1 2 3 4; do
    Rscript scripts/validation/run_chain.R $spec $c &
  done
done
wait
done
Rscript scripts/validation/05_convergence.R          # halt if R-hat > 1.05
Rscript scripts/validation/02_compute_held_out_loglik.R
Rscript scripts/validation/04_per_cancer_breakdown.R
Rscript scripts/validation/03_bootstrap_ci.R

```

Software: R 4.6.0 with `posterior` 1.7.0 + the upstream BSFP dependencies; Python 3.12 with `pandas` 2.3.3, `scipy` 1.17.1, `numpy` 2.4.4. The `infinitefactor` package was archived from CRAN on 2025-12-25 and the upstream code uses it indirectly through BSFP; a pure-R port of the relevant functions is documented in the validation pipeline if needed.

Random seeds are fixed throughout: chains 1–4 use seeds 20260509–20260512; the held-out split uses seed 42; bootstrap uses 20260510; permutation nulls use 20260509 and 20260511.

Paper 2 — operator-composition extension

A separate pre-registration covers a methodology-extension question: do operator compositions (combinations of structural priors across hierarchies) recover patterns single-operator priors miss? The specification list (3 covariates × 3 composition rules × 2 hierarchy pairs = 18 Gibbs specs), decision rules, and Bonferroni-corrected validation criteria are locked at `PRE_REGISTRATION_PAPER2.md`.

Compute has run. All 18 specs fit (4 chains × 100k iters each, 18/18 converged at $R\text{-hat} \leq 1.05$). Per-spec verdicts under the pre-registered decision rule ($\geq +2$ nats Δ vs Paper 1 anchor, CI excludes 0,

Bonferroni $p < 0.00278$) are reported in `EXTENSION_VALIDATION_RESULTS.md` . The secondary diagnostic anchor (vs baseline-on-train, no structural priors) is included in the same table.

Paper 3 — MOFA-FLEX niche residual class (cross-substrate test)

A cross-substrate falsification of the same partial-pooling-on-residual-classes methodology, applied to a fundamentally different model family: the MOFA-FLEX joint factor model on paired Xenium (spatial transcriptomics) and Chromium (single-cell) breast-cancer reference data. The residual-class candidate is **niche** (3-category: tumor / stroma / immune), tested via joint refit of the W loading matrix with a hierarchical-niche prior. Pre-registered in `PRE_REGISTRATION_PAPER3_MOFAFLEX_NICHE.md` .

Compute has run. 4 seeds $\times$ 4 specs (SANITY / PRIMARY / SECONDARY / NULL), joint VI refit on cuda. Three of four seeds fully converged; seed 2511021635 SANITY OOM'd twice in anndata sparse merge and was excluded. Finalize aggregated on $n=3$ seeds. Per-cell paired- Δ vs same-seed SANITY:

spec	n_seeds	mean paired Δ (nats/cell)	sd	range
PRIMARY ($\lambda=1$, niche_3cat)	3	-0.84	1.22	[-1.78, +0.53]
SECONDARY ($\lambda=1$, niche_5cat)	3	-0.31	0.93	[-0.91, +0.76]
NULL ($\lambda=1$, shuffled niche_3cat)	3	-0.63	1.93	[-2.84, +0.74]

Pre-registered decision rule (PRIMARY paired $\Delta > +0.5$ nats/cell with CI excluding 0 AND beating NULL by $z \geq 2$): **FALSIFIED**. PRIMARY mean is negative; NULL paired- Δ is comparable in magnitude (and wider in spread). The niche operator class is insufficient on this substrate. See `MOFAFLEX_NICHE_JOINT_REFIT_RESULTS.md` .

Substrate-boundary interpretation: the methodology that succeeded on the Lock 2022 spike-and-slab pan-cancer model (Paper 1, 3-cov MATCHES with parameter reduction post-disposition; 4-cov sensitivity +6.26 nats) does not transfer to a MOFA-FLEX-style joint factor model with niche-as-residual-class. The two substrates differ in (i) loss landscape (per-patient survival likelihood vs per-cell Gaussian VI ELBO), (ii) parameter density (per-cancer betas vs per-factor W rows), and (iii) baseline drift sensitivity (Gibbs posteriors vs VI re-init drift). One or more of these distinctions appears to break the structural-prior mechanism in this direction.

The $n=3$ caveat is real: seed 1 SANITY's OOM was substrate-specific (anndata sparse merge alloc at ~3.6 GB), not a methodology failure. The 3-seed result is internally consistent across seeds 1636/1637/1638 (NULL noise dominates class signal in all three).

Caveats

- **Validation tests three pre-specified covariates (post-disposition; four in the 4-cov sensitivity)** out of 68. The result does not generalize to the full covariate set. The 3-cov $\leftrightarrow$ 4-cov

disposition trail is in [LOCK_DIAGNOSTIC_DISPOSITION.md](#).

- **Joint validation can mask a noise-tier covariate** carrying the margin past significance. The pre-registered marginal-contribution diagnostic (CHECK 1 + CHECK 2) is the reason cov 10.1 is dropped from the headline. Forward replications should pre-register both the joint and marginal tests.
- **Single 80/20 split**, not k-fold cross-validation. Tighter CIs and per-cancer subgroup tests would require k-fold.
- **Tissue origin** is not used in any validation specification. It escapes the BIDIFAC+ block-presence confound but only marginally beats the granularity-nesting null. A tissue-origin spec would need a separate pre-registration and held-out test.
- **KIRP and STAD** lose under both validation specifications. The gain is heterogeneous; understanding why these cancers worsen is a candidate follow-up.
- **n=29** is small. The screening was demonstrably underpowered for 319 simultaneous tests at Bonferroni rigor. The validation succeeded by testing joint predictive contribution rather than per-cell significance.

Citing

This repository (provisional):

Humphrey, N. (2026). Pan-Cancer Hierarchical Spike-and-Slab — Replication and Structural-Prior Validation. GitHub: [URL TBD]. Zenodo: [DOI TBD].

The replicated method:

Samorodnitsky, S., Hoadley, K. A., & Lock, E. F. (2022). A Hierarchical Spike-and-Slab Model for Pan-Cancer Survival Using Pan-Omic Data. BMC Bioinformatics 23(1):235.

The BIDIFAC+ inputs:

Park, J., Lock, E. F., & Hoadley, K. A. (2022). Bidimensional Linked Matrix Factorization for Pan-Omics Pan-Cancer Analysis. Annals of Applied Statistics 16(1):484-501.

License

MIT. The replication-baseline scripts in `scripts/replication/` are patched verbatim from the upstream repository. Patches to upstream files are documented inline in source. See `LICENSE`.